



C10CF Party + Hackdockingfest

Oktober 1st, 2025 1800 CEST
JobRad GmbH, Freiburg im Breisgau



10 Years of CNCF!





3 Years of⁺ Dock Meetup

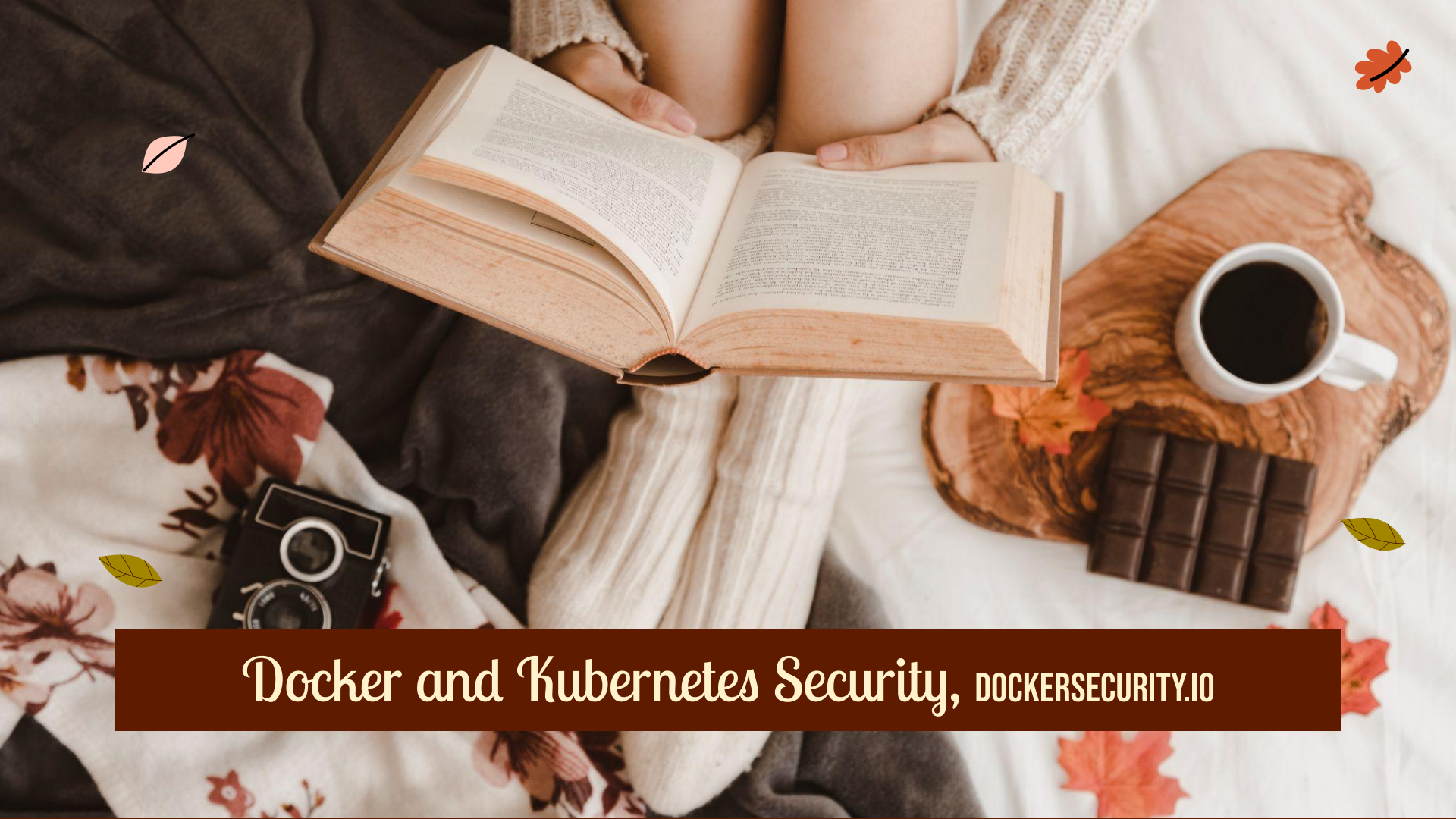
[DOCKBURG.COM](https://dockburg.com)



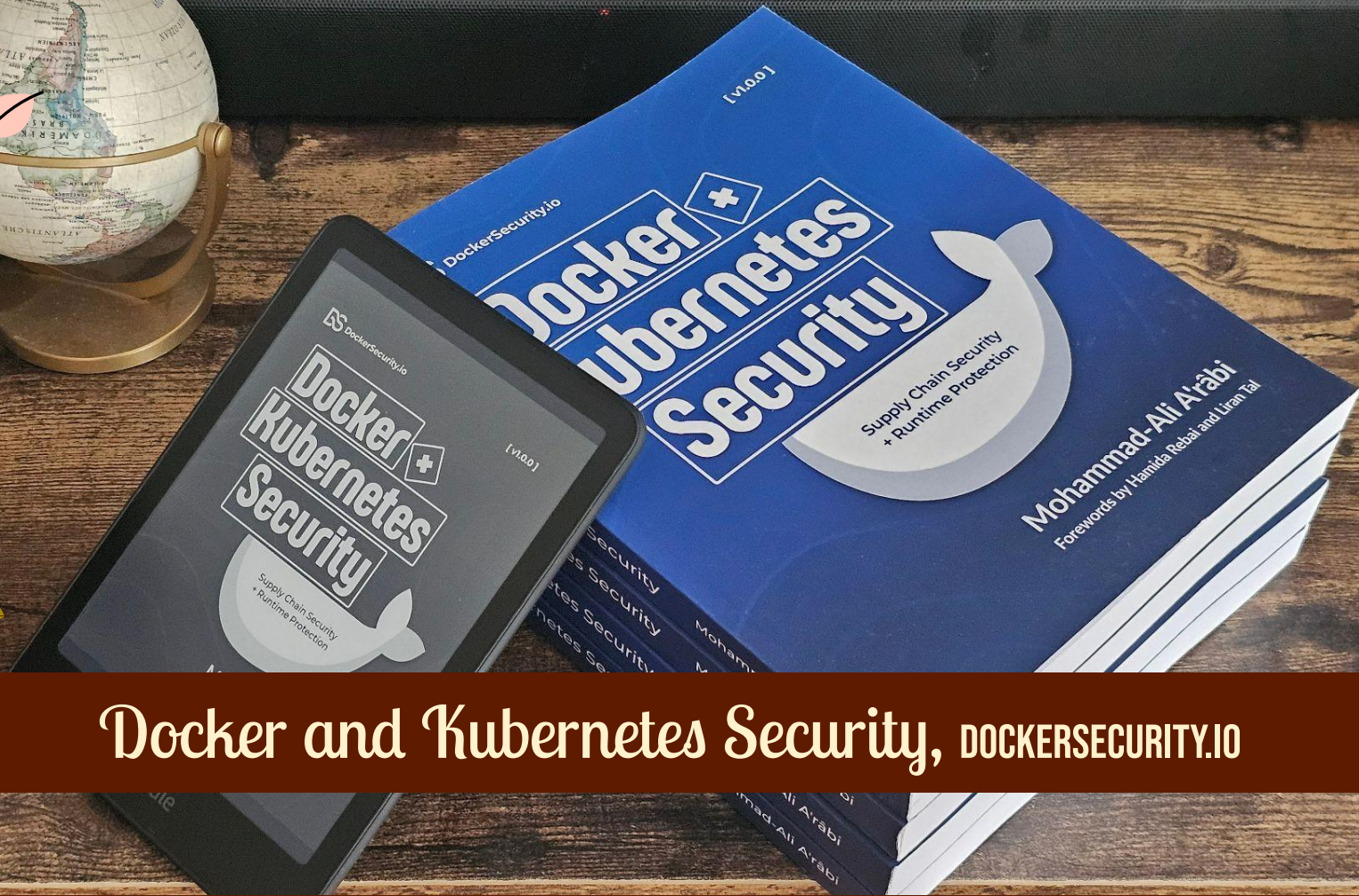


Hackdöcker- fest ist hier!

October 11th, 2022 18:00 CEST
Recyda GmbH, Freiburg



Docker and Kubernetes Security, [DOCKERSECURITY.IO](https://dockersecurity.io)



Docker and Kubernetes Security, DOCKERSECURITY.IO

Docker Commandos

- BAKE A DOCKER CAKE @ PLATFORMCON
- DOCKER DEEP DIVE @ WEAREDEVELOPERS
- 10 DOCKER COMMANDOS @ JOBRAD





Docker Deep Dive

~~to~~ 5 *Docker Kommandos*

You Probably Didn't Know About

~~September 11th ⇒ 10:00 ⇒ Oslo~~

September 1st ⇒ 18:00 ⇒ Oslo



DOCKER COMMANDOS



Workshop Links

GitHub Repo:

<https://github.com/DockerSecurity-io/commandos>

Article:

How to Containerize a Java Application Securely



Basic Docker Commands

[cnt = container]

`docker version` - Show Docker version
`docker ps` - List running containers
`docker image ls` - List local images
`docker run <image>` - Run cnt in detached mode
`docker attach <container>` - Attach to running cnt
`docker exec <container> <cmd>` - Run a new process on a running cnt
`docker stop <container>` - Stop a running cnt
`docker logs <container>` - Show logs of a cnt
`docker rm <container>` - Remove a cnt
`docker rmi <image>` - Remove an image

Docker Build / Buildx

`docker build -t <tag> <context>` - Build image
`docker build -t <tag> .` - Build assuming Dockerfile is in the current directory and Docker should have access to whole tree to build
`docker build -f <dockerfile-path> <context>` - Build image based on a Dockerfile and context
`docker build -f Dockerfile` - No context build
`docker buildx build --platform linux/amd64,linux/arm64 .` - Multi-arch build
`docker buildx create --use --name mybuilder` - Create and switch to a new builder
`docker buildx inspect` - Show current builder info and supported platforms
`docker buildx use <builder>` - Switch to an existing builder instance

This cheatsheet belongs to:

Basic Docker Compose Commands

`docker compose up` - Start services (attached)
`docker compose up -d` - Start services (detached)
`docker compose build` - Build or rebuild services
`docker compose down` - Stop all services
`docker compose logs -f` - Follow logs from services
`docker compose ps` - List status of all services
`docker compose exec <service> <cmd>` - Run command in a running service container
`docker compose stop <service>` - Stop a service
`docker compose config` - Validate and view final config

Secure Dockerfile Tips

- ✓ Use multi-stage builds
- ✓ Run as non-root user (e.g. USER appuser)
- ✓ Use minimal base images (e.g. distroless, alpine)
- ✓ Avoid installing unnecessary tools
- ✓ Pin package versions and digest your base images
- ✓ Generate and attest SBOMs

Static C++ Build with SBOM

`ARG BUILDKIT_SBOM_SCAN_STAGE=true` - Enable SBOM scan for all stages upon build

`FROM ubuntu:latest AS build` - Build stage
`RUN apt-get update && apt-get install -y build-essential` - Install g++
`COPY hello.cpp .` - Copy source file
`RUN g++ -o hello hello.cpp -static` - Compile statically

`FROM scratch` - Distroless (no base image)
`COPY --from=build /app/hello /hello` - Copy only the binary
`CMD ["/hello"]` - Run the statically built binary

Used to demonstrate SBOM scanning across build stages using BuildKit. Without the initial flag, the generated SBOM will be empty.

Learn more:

<https://docs.docker.com/guides/cpp/security/>

Workshop-Specific Commands

`docker init` - Scaffold Dockerfile, Compose, etc.
`docker sbom <image>` - Extract and show SBOM
`docker sbom --format spdx > sbom.json` - Export SBOM with SPDX format
`docker buildx bake` - Build Docker images based on configurations in docker-bake.hcl
`docker buildx build --attest type=sbom`
`--push` - Build Docker image and generate SBOM, then push with SBOM attestations
`docker scout cves <image>` - Check image for CVEs
`docker debug <container>` - Inspect the container
`docker debug <image>` - Inspect the Docker image
`docker model list` - List the available LLMs locally
`docker model run <model>` - Run LLMs with Docker
`docker ai "Review my Dockerfile"` - Ask Gordon

Bonus / Advanced Commands

`docker context ls` - List all Docker contexts
`docker scout recommendations <image>` - Suggest security improvements
`docker buildx imagetools inspect <image>`
`--format "{{ json .SBOM.SPDX }}"` - View SBOM attestation for the image from registry
`docker system prune` - Remove unused containers, images, networks, etc.

Useful Links

git-weekly.com
Blog on Git and Docker



DockerSecurity.io
Docker Security book



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GitHub: <https://github.com/aerabi/docker-cheatsheet>
version 1.0.1 - September 10, 2025

01. Docker Init



Docker Init is a way to Dockerize your project quickly.

- Available on Docker Desktop 4.27 or later, GA
- **Command:** `docker init`
- **Generates:** Dockerfile, compose.yaml, .dockerignore, README.Docker.md
- **Supports:** Go, Python, Java, Node, Rust, ASP.NET, Rust, and PHP



02. Docker Bake

Docker Bake is to Docker build, what Docker Compose is to Docker run.

- Available on Docker CE and Docker Desktop, GA
- **Command:** `docker buildx bake`
- **Config File:** HCL, YAML, or JSON: `docker-bake.hcl`



02. Docker Bake

```
group "default" {  
  targets = ["frontend", "backend"]  
}  
  
target "frontend" {  
  context = "./frontend"  
  dockerfile = "frontend.Dockerfile"  
  args = {  
    NODE_VERSION = "22"  
  }  
  tags = ["myapp/frontend:latest"]  
}  
  
target "backend" {  
  context = "./backend"  
  dockerfile = "backend.Dockerfile"  
  args = {  
    GO_VERSION = "1.23"  
  }  
  tags = ["myapp/backend:latest"]  
}
```



03. Docker SBOM

Docker SBOM is a command to generate Software Bill of Materials out of a Docker image.

- Available on Docker Desktop and installable on Docker CE as a CLI plugin, GA
- **Command:** `docker sbom <image>`
- **Alternative:** Docker Scout has an alternative for this command: `docker scout sbom <image>`
- **SBOM:** SBOM is used for supply-chain security



04. SBOM Attestations

When building a Docker image, you can generate the SBOM and add it to the Docker Hub (or any other registry) as an attestation.

- Available on Docker Desktop and Docker CE, GA
- **Command:** `docker buildx build --sbom=true --push -t gitweekly/git-weekly .`
- **Usage:** Docker Scout checks for SBOM attestations
- **Multi-Stage:** Can include dependencies from build stages
- Check C++ guide: docs.docker.com/guides/cpp/security



05. Docker Scout

Docker Scout is Docker's security solution.

- Available on Docker Desktop and installable as a CLI plugin on Docker CE, GA
- **Command:** `docker scout cves ubuntu:24`
- **Install:** check github.com/docker/scout-cli
- **Usage:** Can be used to check CVEs, recommend solutions for the vulnerabilities, etc.

